

Time: 30 minutes

Max marks: 15

Instructions:

- Do not plagiarize. Do not assist your classmates in plagiarism.
- Define all your variables properly, and show your full solution to get full credit.
- Attempt all questions that you can.
- In the unlikely case a question is not clear, discuss it with an invigilating TA/TF. Please ensure that you clearly include any assumptions you make, even after clarification from the invigilator.

1. (5 points) **Linear Algebra.**

Consider 3D space $\mathbb{R}^3$. We want to project any arbitrary vector $\mathbf{v} = [v_x, v_y, v_z]^T$ orthogonally onto the XY -plane.

- (a) ($\frac{1}{2}$ point) Write down the 3×3 matrix $\mathbf{P}_{xy}$ that performs this transformation such that $\mathbf{v}' = \mathbf{P}_{xy}\mathbf{v}$ lies fully in the XY -plane.
- (b) (1 point) Verify that $\mathbf{P}_{xy}$ satisfies the two defining algebraic properties of an orthogonal projection matrix:
 - i. ($\frac{1}{2}$ point) **Idempotence:** $\mathbf{P}_{xy}^2 = \mathbf{P}_{xy}$
 - ii. ($\frac{1}{2}$ point) **Symmetry:** $\mathbf{P}_{xy}^T = \mathbf{P}_{xy}$
- (c) ($\frac{1}{2}$ point) Identify the unit vector $\hat{\mathbf{p}}$ such that $\mathbf{P}_{xy} = \mathbf{I} - \hat{\mathbf{p}}\hat{\mathbf{p}}^T$, where $\mathbf{I}$ is the identity matrix of appropriate dimension. What is the null space of this matrix $\mathbf{P}_{xy}$?
- (d) (3 points) Given a set of orthonormal vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$, $\mathbf{u}_i \in \mathbb{R}^d, i = \{1, \dots, k\}$, $k \leq d$, and a matrix $\mathbf{U}_{d \times k}$ that has $\mathbf{u}_i, i = \{1, \dots, k\}$ as columns,
 - i. (1 point) Construct a $d \times d$ projection matrix $\mathbf{P}_{U'}$ that performs an orthogonal projection of a vector $\mathbf{v} \in \mathbb{R}^d$ to the null space of $\mathbf{U}^T$ ¹.
 - ii. ($\frac{1}{2}$ point) What is the rank of $\mathbf{P}_{U'}$?
 - iii. (1 point) Construct a $d \times d$ projection matrix $\mathbf{P}_U$ that performs an orthogonal projection of a vector $\mathbf{v} \in \mathbb{R}^d$ to the column space of $\mathbf{U}$.
 - iv. ($\frac{1}{2}$ point) What is the rank of $\mathbf{P}_U$?

{Hint: Recall that the null space of a matrix $\mathbf{A}$ is defined as $N(\mathbf{A}) = \{\mathbf{x} | \mathbf{A}\mathbf{x} = \mathbf{0}\}$, where we assume $\mathbf{A}$ and $\mathbf{x}$ are of appropriate dimensions. Similarly, the column space of a matrix $\mathbf{A}$ is defined as the span of all column vectors of $\mathbf{A}$.}

Solution:

- (a) The matrix $\mathbf{P}_{xy}$ maps $(x, y, z) \mapsto (x, y, 0)$. By inspection:

$$\mathbf{P}_{xy} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Indeed,

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ 0 \end{bmatrix}$$

¹This transpose was unfortunately missing in the question paper. The null space of $\mathbf{U}$ would actually be 0, given the k orthogonal vectors.

(b) i. **Idempotence:**

$$P_{xy}^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = P_{xy}$$

ii. **Symmetry:** Transposing a diagonal matrix leaves it unchanged:

$$P_{xy}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = P_{xy}$$

(c) The vector $\hat{\mathbf{p}} = \hat{\mathbf{z}} = [0, 0, 1]^T$ leads to the following projection matrix:

$$\begin{aligned} \mathbf{P}_{xy} &= \mathbf{I} - \hat{\mathbf{p}}\hat{\mathbf{p}}^T \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Since $\mathbf{P}_{xy} = \mathbf{I} - \hat{\mathbf{p}}\hat{\mathbf{p}}^T$ annihilates precisely the component of $\mathbf{v}$ along $\hat{\mathbf{p}}$, its null space is given by the span of $\hat{\mathbf{p}}$:

$$N(\mathbf{P}_{xy}) = \text{span}\{\hat{\mathbf{p}}\} = \text{span}\left\{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}\right\}$$

This corresponds to the z -axis. Any vector of the form $[0, 0, z]^T$ satisfies $\mathbf{P}_{xy}[0, 0, z]^T = \mathbf{0}$, confirming that it lies in the null space.

(d) Let $\mathbf{U} = \mathbf{U}_{d \times k}$.

- i. $\mathbf{P}_{U'} = \mathbf{I}_{d \times d} - \mathbf{U}\mathbf{U}^T$ is the projection matrix that projects a d -dimensional vector to the null space of $\mathbf{U}^T$.
The rubric should correctly mark responses that identify the error in the question or use a projection matrix that maps to the zero vector.
- ii. The rank of $\mathbf{P}_{U'}$ is $d - k$, as it projects a d -dimensional vector to the orthogonal complement of the k -dimensional column space spanned by $\mathbf{u}_i$, $i = 1, \dots, k$.
- iii. $\mathbf{P}_U = \mathbf{U}\mathbf{U}^T$ is the projection matrix that projects $\mathbf{v} \in \mathbb{R}^d$ to $\text{Col}(\mathbf{U}_{d \times k})$.
- iv. The rank of $\mathbf{P}_U$ is k , the same as the rank of $\mathbf{U}$.

2. (5 points) Probability and Statistics - Continuous.

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ be a univariate continuous random variable following a Gaussian distribution with mean $\mu \in \mathbb{R}$ and variance $\sigma^2 > 0$.

(a) (2 points) Let $Y = aX$ where $a \in \mathbb{R} \setminus \{0\}$ is a constant scalar.

- i. (1 point) Using properties of expectation and variance, determine the mean μ_Y and variance σ_Y^2 of Y .
- ii. (1 point) Derive the probability density function (PDF) $f_Y(y)$ in terms of μ, σ^2 and a .
- (b) (2 points) Let $Z = aX + b$ where $a, b \in \mathbb{R}$ with $a \neq 0$.
 - i. (1 point) Find the mean μ_Z and variance σ_Z^2 of Z .
 - ii. (1 point) What choice of a and b transforms X into a standard normal variable $Z \sim \mathcal{N}(0, 1)$?
- (c) (1 point) Let $\mathbf{X} \in \mathbb{R}^2$ be a multivariate Gaussian random vector denoted as $\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}_X, \boldsymbol{\Sigma}_X)$, where $\boldsymbol{\mu}_X \in \mathbb{R}^2$ is the mean vector and $\boldsymbol{\Sigma}_X \in \mathbb{R}^{2 \times 2}$ is a symmetric positive-definite covariance matrix.

Consider the transformation:

$$\mathbf{Y} = \mathbf{A}\mathbf{X}$$

where $\mathbf{A} \in \mathbb{R}^{2 \times 2}$ is full rank.

53 Using vector expectation and covariance matrix definitions, derive the mean vector $\boldsymbol{\mu}_Y \in \mathbb{R}^2$ and covariance matrix $\boldsymbol{\Sigma}_Y \in \mathbb{R}^{2 \times 2}$ of $\mathbf{Y}$ in terms of $\mathbf{A}, \boldsymbol{\mu}_X$ and $\boldsymbol{\Sigma}_X$. To get full credit, write the expression in matrix form.

{ *Hint:* The PDF of the normal distribution $\mathcal{N}(\mu, \sigma^2)$ is given as

$$f_X(x; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

}

Solution:

- (a) i. Using linearity of expectation and properties of variance:

$$\mu_Y = \mathbb{E}[Y] = \mathbb{E}[aX] = a\mathbb{E}[X] = a\mu$$

$$\sigma_Y^2 = \text{Var}(Y) = \text{Var}(aX) = a^2 \text{Var}(X) = a^2 \sigma^2$$

Thus, $Y \sim \mathcal{N}(a\mu, a^2\sigma^2)$.

- ii. Let $g(x) = ax$, so $y = g(x)$ and $x = g^{-1}(y) = \frac{y}{a}$. The transformation derivative is $\frac{dx}{dy} = \frac{1}{a}$. By the change-of-variables formula for continuous density functions:

$$f_Y(y) = f_X(g^{-1}(y)) \cdot \left| \frac{d}{dy} g^{-1}(y) \right| = f_X\left(\frac{y}{a}\right) \cdot \frac{1}{|a|}$$

Substituting the univariate normal PDF $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$:

$$f_Y(y) = \frac{1}{|a|\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y/a - \mu)^2}{2\sigma^2}\right) = \frac{1}{\sqrt{2\pi(a^2\sigma^2)}} \exp\left(-\frac{(y - a\mu)^2}{2(a^2\sigma^2)}\right)$$

This confirms Y is Gaussian with mean $a\mu$ and variance $a^2\sigma^2$.

- (b) i. For the affine transformation $Z = aX + b$:

$$\mu_Z = \mathbb{E}[aX + b] = a\mathbb{E}[X] + b = a\mu + b$$

$$\sigma_Z^2 = \text{Var}(aX + b) = \text{Var}(aX) = a^2\sigma^2$$

- ii. Since shifting a distribution by a constant b does not change its functional normal form (it translates the center), $Z \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$. Its density is:

$$f_Z(z) = \frac{1}{\sqrt{2\pi a^2 \sigma^2}} \exp\left(-\frac{(z - (a\mu + b))^2}{2a^2 \sigma^2}\right)$$

- iii. To obtain $\mu_Z = 0$ and $\sigma_Z^2 = 1$, set $a\mu + b = 0$ and $a^2\sigma^2 = 1$. Choosing $a = \frac{1}{\sigma}$ and solving $b = -\frac{\mu}{\sigma}$ gives:

$$Z = \frac{X - \mu}{\sigma} \sim \mathcal{N}(0, 1)$$

This is the standard score (z-score) normalization formula.

(c) **Mean vector:**

$$\mu_Y = \mathbb{E}[Y] = \mathbb{E}[AX] = A\mathbb{E}[X] = A\mu$$

Covariance matrix: By definition, $\Sigma_Y = \mathbb{E}[(Y - \mu_Y)(Y - \mu_Y)^T]$. Noting that $Y - \mu_Y = (AX) - (A\mu) = A(X - \mu)$:

$$\Sigma_Y = \mathbb{E}[A(X - \mu)(A(X - \mu))^T] = A\mathbb{E}[(X - \mu)(X - \mu)^T]A^T = A\Sigma A^T$$

Thus, $Y \sim \mathcal{N}(A\mu, A\Sigma A^T)$.

3. (4 points) **Probability and Statistics - Discrete.**

A biased coin has a probability $p = 0.7$ of heads ($X = 1$) and $1 - p = 0.3$ of tails ($X = 0$).

- (1 point) Write down the complete Probability Mass Function (PMF) of X .
- (1 point) Calculate the expectation $\mathbb{E}[X]$ and the variance $\text{Var}(X)$ using their respective definitions. (**Note:** no credit for simply writing the formula for the mean and variance of the Bernoulli distribution.)
- (2 points) Suppose you toss this same coin twice independently, yielding two i.i.d. Bernoulli random variables X_1 and X_2 . Given that at least one flip led to heads ($X_1 + X_2 \geq 1$), calculate the conditional probability that both flips landed on heads ($X_1 = 1$ and $X_2 = 1$).

Solution:

- (a) i. The PMF of X is given as

$$P(X = x) = \begin{cases} 0.7, & \text{if } x = 1 \\ 0.3, & \text{if } x = 0 \\ 0, & \text{otherwise} \end{cases}$$

- ii. Expectation:

$$\mathbb{E}[X] = \sum x \cdot P(X = x) = (1 \cdot 0.7) + (0 \cdot 0.3) = 0.7$$

Variance:

First, we compute $\mathbb{E}[X^2]$:

$$\mathbb{E}[X^2] = (1^2 \cdot 0.7) + (0^2 \cdot 0.3) = 0.7$$

Then, compute the variance:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 0.7 - (0.7)^2 = 0.7 - 0.49 = 0.21$$

- iii. Let A be the event that both flips are heads ($X_1 = 1, X_2 = 1$), and let B be the event that at least one flip is heads ($X_1 + X_2 \geq 1$).

By the conditional probability formula:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

- Since both flips being heads implies at least one flip is heads, $A \subseteq B$, so $A \cap B = A$:

$$P(A \cap B) = P(X_1 = 1, X_2 = 1) = P(X_1 = 1) \cdot P(X_2 = 1) = 0.7 \times 0.7 = 0.49$$

- $P(B)$ is the event of getting at least one heads, i.e., the complement of getting no heads (both tails):

$$P(B) = 1 - P(X_1 = 0, X_2 = 0) = 1 - (0.3 \times 0.3) = 1 - 0.09 = 0.91$$

Substituting these values yields:

$$P(A | B) = \frac{0.49}{0.91} = \frac{49}{91} = \frac{7}{13} \approx 0.5385$$

4. (1 point) **Vector Differentiation.** Let $\mathbf{x} \in \mathbb{R}^n$ be the input to function $f(\mathbf{x}) = \|\mathbf{x}\|_2^2 = \mathbf{x}^T \mathbf{x} = \sum_{i=1}^n x_i^2$. Write the gradient vector $\nabla_{\mathbf{x}} f$ for this function.

Solution: Since $f(\mathbf{x}) = \sum_{i=1}^n x_i^2$, differentiating with respect to each component:

$$\frac{\partial f}{\partial x_i} = 2x_i, \quad i = 1, \dots, n$$

Stacking these partial derivatives into a vector:

$$\nabla_{\mathbf{x}} f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} = \begin{bmatrix} 2x_1 \\ \vdots \\ 2x_n \end{bmatrix} = 2\mathbf{x}$$

(Equivalently, using the identity $\nabla_{\mathbf{x}}(\mathbf{x}^T A \mathbf{x}) = (A + A^T)\mathbf{x}$ with $A = \mathbf{I}$: $\nabla_{\mathbf{x}} f = 2\mathbf{I}\mathbf{x} = 2\mathbf{x}$.)

Detailed Solution

Definition of Gradient Vector

The gradient of a scalar-valued function $f(\mathbf{x})$ with respect to an n -dimensional input vector $\mathbf{x} = [x_1, x_2, \dots, x_n]^T \in \mathbb{R}^n$ is defined as the column vector of its partial derivatives:

$$\nabla_{\mathbf{x}} f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

Component-wise Partial Differentiation

Expanding the summation representation of the function:

$$f(\mathbf{x}) = x_1^2 + x_2^2 + \cdots + x_i^2 + \cdots + x_n^2$$

Taking the partial derivative of $f(\mathbf{x})$ with respect to a single component x_i ($i \in \{1, 2, \dots, n\}$):

$$\frac{\partial f}{\partial x_i} = \frac{\partial}{\partial x_i} (x_1^2 + x_2^2 + \cdots + x_i^2 + \cdots + x_n^2)$$

Since all terms x_j^2 for $j \neq i$ do not depend on x_i , their partial derivatives equal 0. Applying the standard power rule to x_i^2 :

$$\frac{\partial f}{\partial x_i} = 2x_i \quad \text{for all } i = 1, 2, \dots, n$$

Stacking into Vector Form

Substituting the partial derivative results back into the gradient vector format:

$$\nabla_{\mathbf{x}} f = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} = \begin{bmatrix} 2x_1 \\ 2x_2 \\ \vdots \\ 2x_n \end{bmatrix} = 2 \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = 2\mathbf{x}$$

Alternative Method: Matrix Calculus Identity

Using the general matrix derivative identity for quadratic forms $\nabla_{\mathbf{x}}(\mathbf{x}^T \mathbf{A} \mathbf{x}) = (\mathbf{A} + \mathbf{A}^T)\mathbf{x}$:

Setting $\mathbf{A} = \mathbf{I}$ (where $\mathbf{I}$ is the $n \times n$ identity matrix):

$$\nabla_{\mathbf{x}}(\mathbf{x}^T \mathbf{x}) = (\mathbf{I} + \mathbf{I}^T)\mathbf{x} = 2\mathbf{I}\mathbf{x} = 2\mathbf{x}$$

Grading Rubric (Total: 1 Mark)

Marks	Criteria & Assessment Breakdown	Key Elements to Check
0.5 Mark	Component Partial Derivative Correctly applies differentiation with respect to individual vector elements x_i , recognizing that $j \neq i$ terms evaluate to zero.	$\frac{\partial f}{\partial x_i} = 2x_i$ (or explicit component expansion)
0.5 Mark	Vector Assembly & Final Output Correctly constructs the gradient column vector from partial derivatives or successfully applies matrix calculus identity to arrive at the final vector answer.	$\nabla_{\mathbf{x}} f = 2\mathbf{x}$ (or equivalent column vector)